



Detection of buildings that are suitable for
green roofs using Machine Learning
Artificial Intelligence in Human Water Systems

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1. Topic and Motivation

Investment in green infrastructure serves a critical role in the future of urban areas affected by climate change and thus rising temperatures and water imbalance. Berlin, as one such urban space, has a longstanding history with green roofs and the benefits they provide in heat and water regulation as well as biodiversity.

Beginning in 1839 with the invention of the Tar-Paper-Green roofs (Holzzementdach) and their following increased implementation during the late Gründerzeit, furthered by installations of Modern-Extensive-Green roofs since the 1980s,¹ Berlin is an ideal starting point in researching and assessing the suitability of differing Green Roofs in green infrastructure.

Although there are already 5,4% of all buildings in Berlin equipped with extensive or intensive green roofs we want to research to see if there are more buildings that can be upgraded with a green roof.² This would enhance the positive effects that come with green roofs.

In this study we want to extend the research that has already been done so far. One of the papers investigates the green roof potential using remote sensing data.³ Like in this paper there are multiple approaches detecting the green roof potential using geospatial data like orthophotos in combination with deep learning algorithms.

Since these methods require a deeper understanding of the field of deep learning, we decided to follow an easier approach that uses open access geodata to train a machine learning algorithm. The approach of using only Gis data was also already used in a Study that covered the city of Braunschweig. Next to the suitability of new green roofs, the study focused on the potential impacts of the new green roofs. Since the city of Braunschweig is a lot smaller and therefore has different characteristics, we focus on Berlin in this study.⁴

With this approach we aim to see, whether it is possible to predict the suitability of a roof to be retrofitted with a green roof based on open access and easy to handle data.

2. Background and Data

The study area in this paper is the city of Berlin. As the capital and largest city of Germany it is an urban area with a dense city center and lower dense outskirts.

Berlin's climate is characterized by a transition between maritime and continental influences, making it one of Germany's warmest cities with an annual average temperature of **13.1°C**. A significant "urban heat island" effect exists, where dense city centers remain much warmer than outlying green spaces, particularly during summer nights when temperatures often stay above **20°C**. While summers bring the highest temperatures (averaging **23.6°C**) and peak sunshine, they also see the highest average rainfall.

¹ Manfred Köhler, Philipp H. Poll: Long-term performance of selected old Berlin greenroofs in comparison to younger extensive greenroofs in Berlin. In: Ecological Engineering, Volume 36, Issue 5, 2010, Pages 722-729, ISSN 0925-8574, <https://doi.org/10.1016/j.ecoleng.2009.12.019>.

² Senate Department for Urban Mobility, Transport, Climate Action and the Environment (Berlin). *Environmental Atlas Berlin: 06.11 Green Roofs*. (n.d.). Available at: <https://www.berlin.de/umweltatlas/nutzung/gruendaecher/> [Accessed 27 Mar. 2026].

³ Sun, L., Wang, J., & Zhang, Y. (2025). Optimizing urban green roof potential through multi-source geospatial data integration. *Cities*, 158, 105782. <https://doi.org/10.1016/j.cities.2025.105782>

⁴ Langemeyer, J., et al. (2017). Predicting the green roof potential of existing buildings. *Urban Forestry & Urban Greening*, 21, 152–162. <https://doi.org/10.1016/j.ufug.2017.01.001>

Conversely, winters are defined by grey skies and sharp easterly winds, with January being the coldest month, often dipping below freezing at night.⁵

Taking these climate conditions into account green roofs may have a resilience improving impact on the city. This impact, however, was already part of other research projects and will not be covered here.⁶

2.1 Data source

The data used in this study is coming from multiple sources. The GeoData Portal of the city of Berlin provides various datasets that can be applied on this topic. To create a dataset that the machine Learning algorithms can be applied on, the data listed below is used.

- Solar potential

The Data about the solar potential contains valuable information about the buildings in the study area of Berlin. Since the requirements for roof greening and upgrading a roof with solar panels are highly similar there are multiple attributes that can be reused. First, the solar potential contains information about the slope of the roofs. This attribute is relevant for green roofing due to the maximum inclination on that a green roof can be installed on.

Besides the slope of the roof segments of each building, the shade properties are another important feature in the suitability assessment. Since the vegetation on green roofs is dependent on varying intensity of sunlight for diverse species of plants to grow, the shading must be included in the data selection.

Another feature used from the solar potential dataset is the number of stories of a building. This information is included because it refers to the static properties of a building. Since higher story buildings require higher fire safety measures which include thicker walls the static load capacity of the building also increases. This would also increase the suitability of green roofing which relies on high load capacity.

- Green Roofs Berlin

This dataset contains information about the current distribution of green roofs in the city. It serves as the validation dataset since it marks all green roofs dated in 2020.

Out of the list of features provided in the dataset, especially the column if a green roof exists is of interest. Other details like the total area of a green roof or the partition of building footprint are excluded to only find roofs that can be upgraded and not the potential area of a green roof.

To keep the approach more focused on the methodology we also don't take the kind of green roof into account. If a building is suitable for an extensive or intensive green roof is out of the boundaries of this detection method.

- Districts

⁵ wetter.de. (n.d.). *Klima Berlin*. RTL interactive GmbH. Retrieved March 27, 2026, from <https://www.wetter.de/klima/europa/deutschland/berlin-s99000032.html>

⁶ Shafique, M., Kim, R., & Rafiq, M. (2020). Green roof benefits, opportunities and challenges – A review. *Urban Forestry & Urban Greening*, 52, 126738. <https://doi.org/10.1016/j.ufug.2020.126738>

A shape file of all the districts in Berlin is used to filter the data. Due to the size of the unprocessed data that includes most of the buildings in Berlin we decided to cut all data down to the area of the district “Neukölln”. This makes the visualization of the data in QGIS more accessible.⁷

3. Methodology

To apply the machine learning algorithm on the data described in chapter two we had to preprocess the data first. The goal of the preprocessing is to get an overview of all the attributes and the data quality. All the coding was done using Python (version 3.13.5) in Visual Studio Code and QGIS was used for visualization of geospatial data.

The data was downloaded in a WFS format, so it had to be transformed into a readable datatype. QGIS provides an export tool for this type of task that made it possible to work with the data as shape files in python.

We used the python package GeoPandas to bring all features together in one dataset. Its function spatial join aggregated the features for each building by merging the attributes of buildings (polygons) that intersect. To prevent unintentional merging of buildings (e.g. neighboring buildings with polygon intersections) we used the representative point method. This method uses a central point for each polygon and checks on which Polygon at the other layer it intersects. Using this method makes sufficient merging possible while ignoring imprecise polygon outlines. With this technique we were able to create a single dataset to apply the machine learning algorithm on. To reduce the size of the data set unused attributes were removed and only the ones delivering potential valuable information are kept.

This method worked out for nearly all of the different data sources. One exception was the dataset with solar potential. Since the polygons in the shape file did not represent the building footprint, but rather different segments of the roof based on the slope the data had to be prepared before the merge with all other data. To bring the segmented roofs into a format that matches the other data that are based on building footprints we aggregated them. The aggregation calculates the portion of the roof area that has a slope less than 15°.⁸ This number is then saved to the whole building footprint and can then be merged with the other data.

Since the datasets do not cover every building in Berlin we decided to keep only the buildings that are included in the dataset with green roofs. With this priority we can use every building as validation data. Other missing values are replaced by average values.

After merging all the features together and removing unnecessary features like geometry information the dataset is prepared for further processing. All the features used are listed in the table below.

⁷ Senate Department for Urban Development, Building and Housing. (n.d.). *Geodaten Berlin* [Geodata Berlin]. Berlin.de. Retrieved March 30, 2026, from <https://www.berlin.de/sen/sbw/stadtdaten/geodaten-berlin/>

⁸ Deutsches Institut für Bautechnik (DIBt). (2021). *European Technical Assessment ETA-13/0557: Bauder waterproofing systems for greened roof surfaces – Resistance to root penetration*. Berlin, Germany. [https://www.dibt.de/pdf_storage/2021/ETA-13!0557\(8.04.01-8!20\)e.pdf](https://www.dibt.de/pdf_storage/2021/ETA-13!0557(8.04.01-8!20)e.pdf)

Table 1: Overview of the features after preprocessing

Feature	Meaning
suitable_roof_area	Area of a roof that has a slope of less than 15° in [m ²]
total_roof	Total roof area [m ²]
suitable_share_pct	Percentage of suitable roof area [%]
anzahl_obergeschoss	Amount of building stories
dominant_orientation	The orientation that is faced by most of the roof area
Avg_slope	Average slope of the roof
Bauweise_schluesel	Type of building as id
verschattung_0	Area that is rarely shaded [m ²]
verschattung_1	Area that is sometimes shaded [m ²]
verschattung_2	Area that is mostly shaded [m ²]

The models of choice in this study are a Classification and Regression Tree (CART) and a Random Forest (RF). Since all of features listed are in a numeric or Boolean format and there is validation data available, we decided on these supervised classification and regression methods.

With CART we can get a simple model that is cheap in terms of resources and may be visualized. It also serves as reference to compare the results with the Random Forest.

In Random Forests Results will be averaged values or a majority voting from several decision trees. Although we don't want to go too deep into the theoretical background it's important to name the hyperparameters we adjusted for the model. Further information about the logic and math behind the model can be found in the literature.⁹

To create the model the preprocessed data was split into training data, test data and the target variable was determined.

As target variable we chose the column "gruendach" which is binary:

1 = green roof exists

2 = no green roof.

The split into a training set and a test split was performed with an 80 to 20 % ratio. In the SciKitlearn package are multiple models provided to apply on your data. We chose the DecisionTreeClassifier and set the max depth to 8. With a minimum of 20 samples per leaf overfitting is prevented. The model is then trained on the training data and evaluated on the test data.

To encounter the imbalance in the target variable the target class was balanced. There are below 5 % green roofs out of all buildings. This method increases the cost for a false positive prediction, so the target variable is predicted with higher accuracy.

The Random Forest was created with 100 decision trees. Every tree has a maximum depth of 15 and a minimum sample size of 10 at every leaf. Like the CART the classes are balanced so that the target variable is predicted with higher accuracy.

For the single CART the confusion matrix and statistical measures will be discussed in the next section. The Random Forest gives out similar results than can be interpreted.

⁹ Biau, G., & Scornet, E. (2016). A random forest guided tour. *TEST*, 25(2), 197–227. <https://doi.org/10.1007/s11749-016-0481-7>

To make use of the predicted scores we implemented a classification system that subdivides all buildings into 3 classes. The ranking is based on the prediction probabilities of the Random Forest model. For every building a probability is determined in the Random Forest model how likely it is a green roof. The probability for each building is put together in a ranked list. This value is then classified into one of the following classes by its percentile:

- 2: 90 percentile -> building belongs to the top 10 % in list of buildings
- 1: 65 percentile -> building belongs to the next 25 % in the list of buildings
- 0: below 65 percentile -> building belongs to the lower 65 % in the list of buildings

The applied classification method makes the visualization in QGIS possible.

Given the size of the data and the therefore needed time of the coding and GIS applications we decided to apply the model first on only the district of Neukölln and afterward on the whole city area. This also makes the results comparable.

4. Results

The results of a Random Forest model and a CART are expressed in several statistic measures. Since the CART consists of nodes, leaves and branches it can be visualized. Given the fact that the RF model consists of multiple Decision Trees it's not possible to visualize the result in a single graphic.

4.1 Berlin complete without threshold tuning

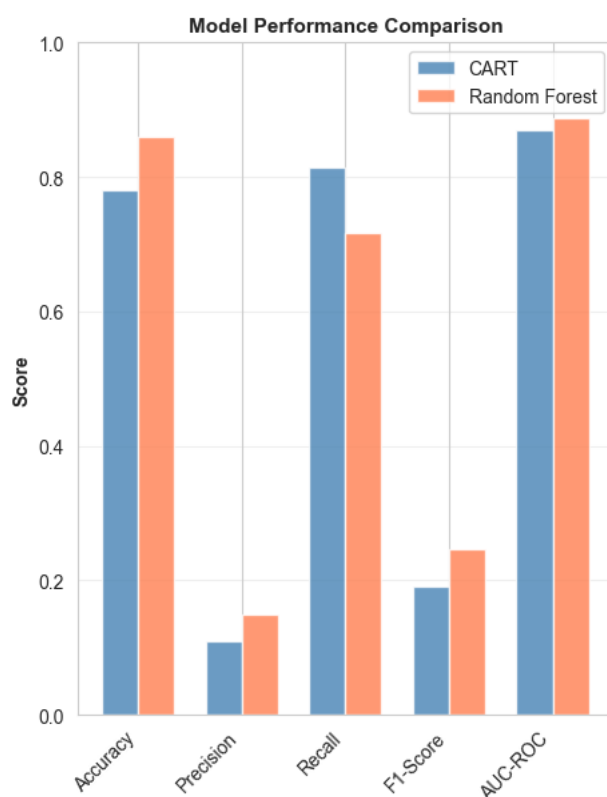


Figure 1: Comparison of basic statistic measurements for CART and RF for the whole city of Berlin

In comparison the Random Forest model performs overall with higher scores in accuracy, precision, F1-Score and the area under curve (AUC-ROC). Only the recall reaches a higher score as shown in **Fehler! Ungültiger Eigenverweis auf Textmarke..**

The parameters can be interpreted as good performance in general. What lowers the quality of the model is the precision and the F1-score. The precision that is defined by

$$precision = \frac{TP}{TP + FP}$$

TP = True Positives, FP = False Positives

shows low values which means that the model performs poorly at predicting the cases in which a green roof exists. This low performance is also reflected by the F1 Score that measures the combination of the other metrics. All the Metrics are calculated from the confusion

matrix. A confusion matrix shows how many samples are classified wrong or right by the model. Like all other metrics the confusion matrix is calculated by the SciKitlearn package. The confusion matrix from the CART and the Random Forest is displayed in **Fehler! Verweisquelle konnte nicht gefunden werden..** On the x-axis is the predicted value and on the y-axis are the actual values. Besides the values on the main diagonal that are the correct classi-

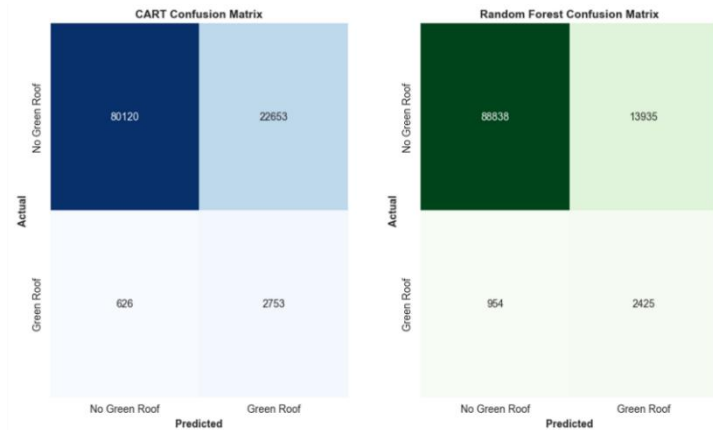


Figure 2: Confusion Matrix of CART and Random Forest for the city of Berlin

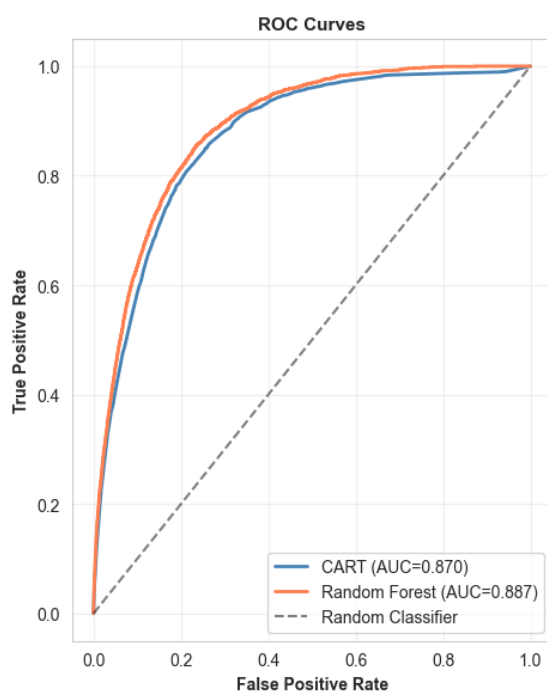


Figure 4: ROC-AUC metric for Random Forest und CART in Berlin

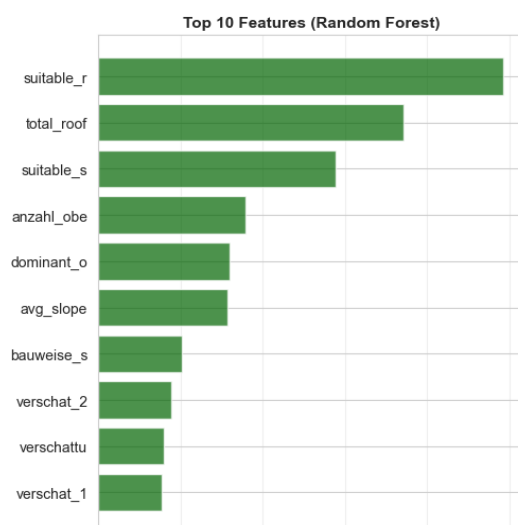


Figure 3: Feature importance for Random Forest

fied values the false positives are noticeable. The high number shows that the model classifies many buildings without green roofs as buildings with green roofs.

Another important metric is the receiver operating characteristic area under curve (ROC-AUC). The plot shows the ratio of the true positive rate and false positive rate. Since the dotted line with a ROC of 0.5 can be interpreted as a random guess, our result show that the model can differentiate between buildings with and without a green roof. Although the model yielded low precision, the high AUC-ROC scores (RF: 0.887, CART: 0.870) demonstrate that the models possess high discriminative power. This confirms that the models effectively learned the underlying features of suitable roofs. The features used for the prediction can also be ranked by their importance for the model. The feature importance is ranked in Figure 6. For each split in a decision tree, the model evaluates how much the chosen feature reduces the node impurity

(here: Gini impurity), weighted by the number of samples in the child nodes. This impurity reduction is accumulated for every feature across all splits where it is used. In the Random Forest, these accumulated values are then averaged over all trees and finally normalized so that the importances sum to one. Consequently, a feature with a high importance value contributes strongly and consistently to separating buildings with and without green roofs, whereas features with low importance play only a minor role in the model's decisions. In our model the suitable roof area is the most important feature followed by the total roof area and the percentage of the suitable roof area. The number of stories of a building also plays an important role as decision criteria.

4.2 Focus to district Neukölln with threshold tuning

To reduce the model size and simplify manual adjustments in the second phase, a Random Forest classifier was applied specifically to the Neukölln district of Berlin. This targeted approach streamlined the analysis and avoided data overload, ensuring the results remained manageable and easier to interpret. The Random Forest classifier was trained on 10,668 roof samples to predict green roof suitability in Berlin-Neukölln under strong class imbalance (275 suitable vs. 10,393 non-suitable). The model was evaluated against a baseline and a Decision Tree benchmark, and overall performance has been assessed with balanced accuracy, F1-score, precision, recall, and confusion matrices. Threshold tuning was applied as a lightweight post-processing step (without retraining): instead of using the default 0.50 cutoff, we tested values from 0.05 to 0.95 and selected 0.25, which gave the best class-1 F1-score.¹⁰ This threshold choice improved minority-class detection and aligned with the project goal of identifying as many plausible opportunities as possible in an early planning stage. At the selected threshold, the model reached a balanced accuracy of 0.62 and increased class-1 recall from 0.11 (at 0.50) to 0.25, while precision for class 1 was 0.33. The resulting class-1 F1-score increased by about 55% (0.283 vs. 0.183), showing a better trade-off between finding true opportunities and limiting false alarms. The confusion matrix supports this interpretation: 10,250 true negatives, 143 false positives, 206 false negatives, and 69 true positives. In practical terms, this means the model keeps high specificity while surfacing substantially more candidate roofs for follow-up assessment than a conservative default setting would provide.

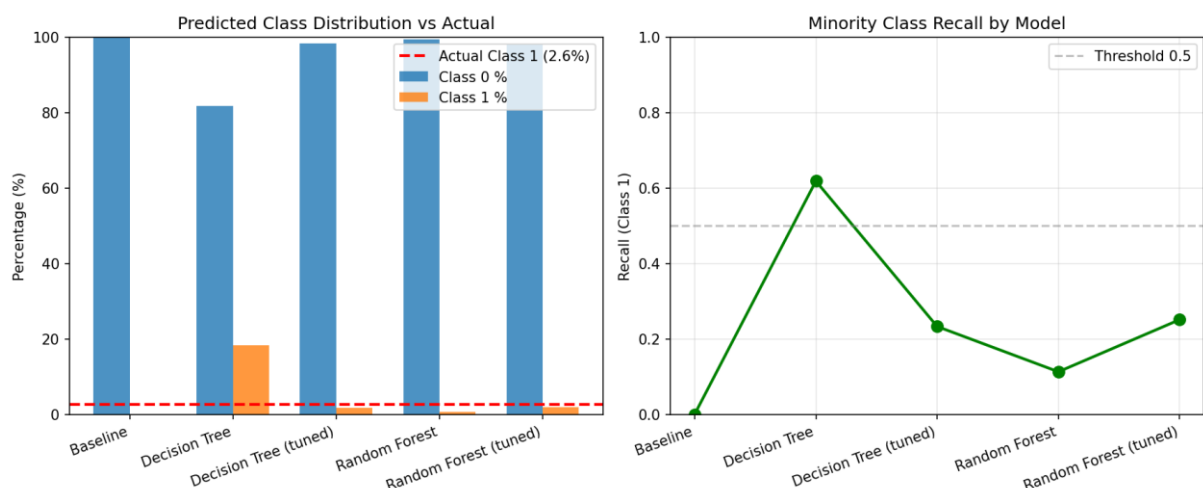


Figure 5: class balance analysis and minority class recall

Feature analysis also adds interpretability: the top-15 feature importance ranking highlights which predictors most strongly drive suitability decisions and helps connect model behavior to domain logic. From an efficiency perspective, the workflow is lightweight and scalable: training requires only about 3-5 seconds and modest computational resources, making repeated runs and periodic updates feasible.

¹⁰ Scikit-learn developers. (n.d.). *Classifier profile: Tuning the decision threshold*. Scikit-learn. https://scikit-learn.org/stable/modules/classification_threshold.html

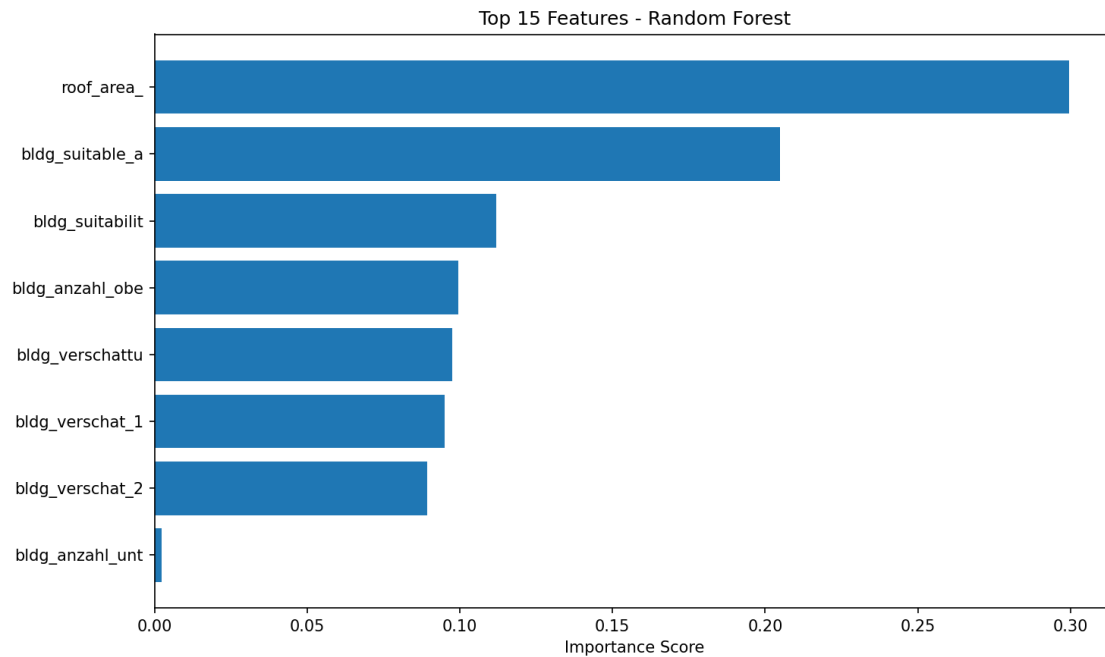


Figure 6: Feature Importance and balancing analysis

For spatial validation, model outputs were visualized in QGIS and compared qualitatively with Google Earth imagery to assess whether predicted hotspots align with real-world roof patterns.

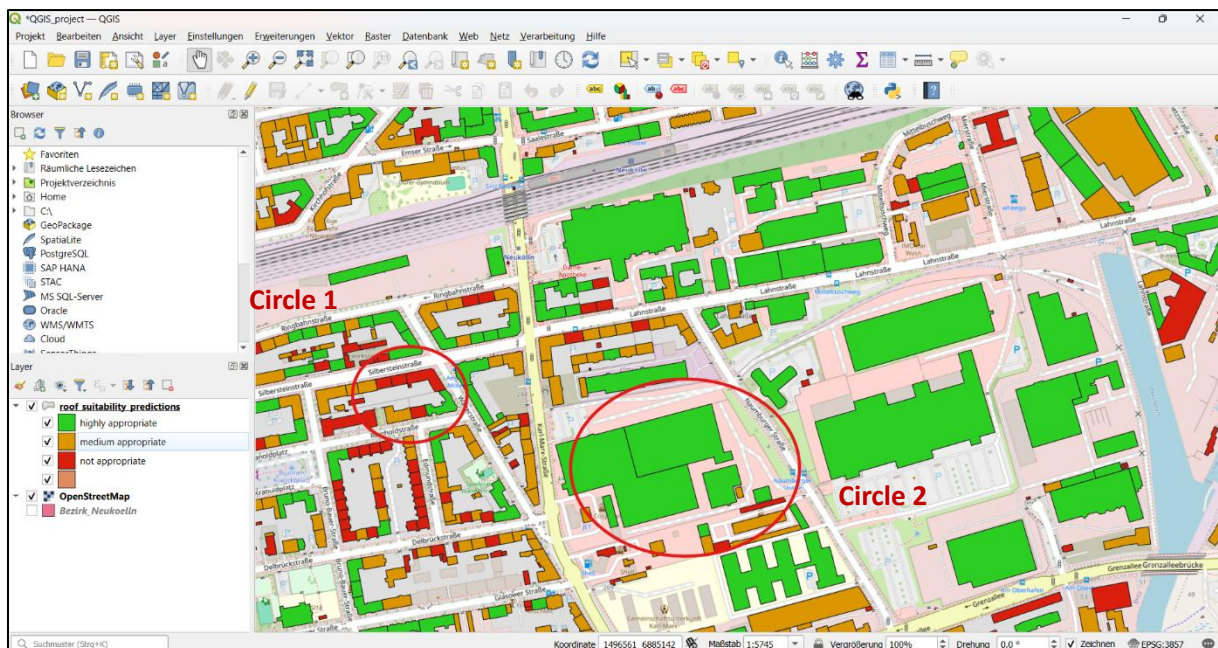


Figure 7: Classification of suitability for green roofs in QGIS (Berlin-Neukölln)

The initial predictions align well with practical civil engineering expectations. As anticipated, industrial facilities and large building blocks show the highest prevalence of green roofs, likely due to their expansive, flat surfaces (Circle 2). Medium-sized structures also show potential, particularly those with high solar exposure and flat roof profiles. However, older residential or heritage buildings remain less suitable than modern industrial halls, consistent with our initial structural assessments (Circle 1).

In comparison with *Google Earth*:

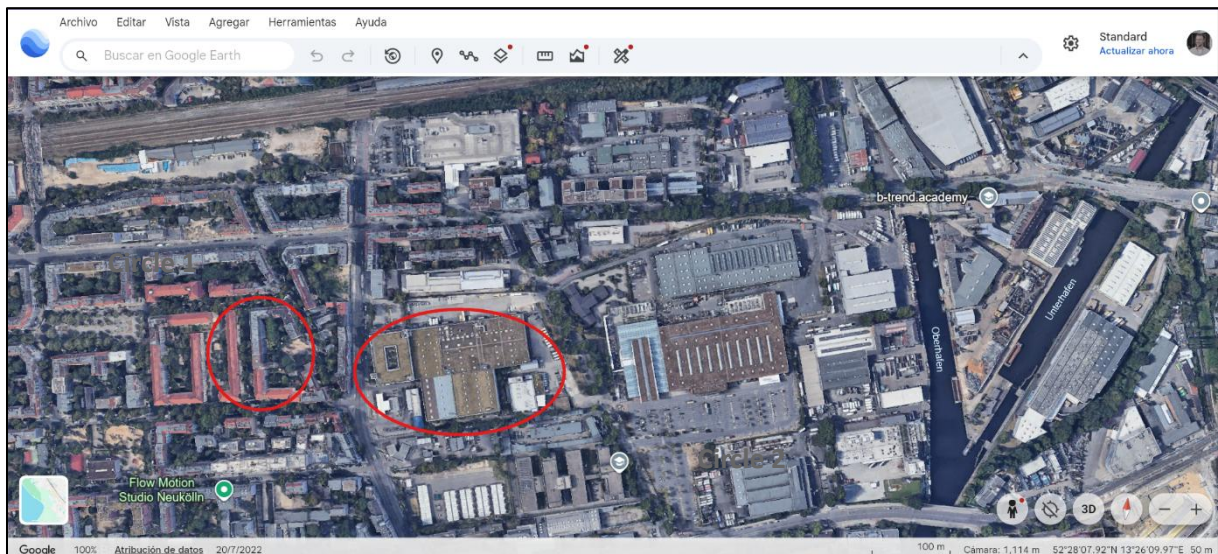


Figure 8: orthophoto of industrial area in Neukölln showing appropriate green roofs (Google Earth: 52°27'55.34"N 13°27'16.56"E)

However, some limitations in precision remain evident. For instance, the Random Forest model classified a church as highly suitable, despite the significant structural and regulatory challenges associated with retrofitting historical buildings.

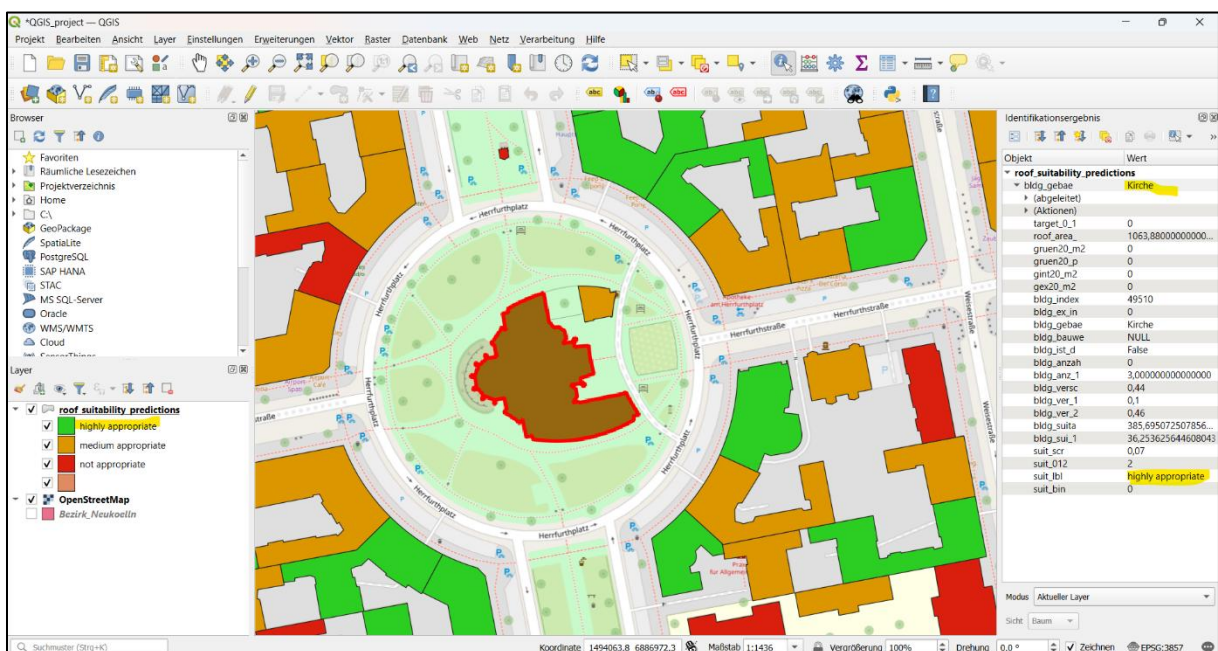


Figure 9: Church predicted as highly appropriate (in QGIS)

In comparison with *Google Earth*:

Detection of buildings that are suitable for green roofs using Machine Learning

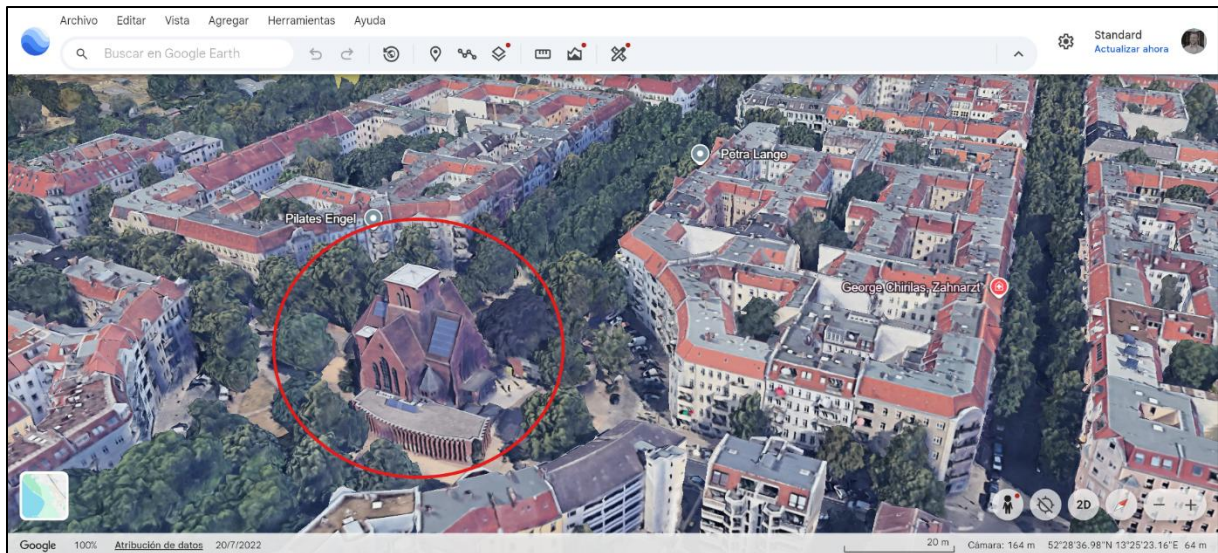


Figure 10: Church in Berlin-Neukölln (Google Earth: 52°28'38.53"N 13°25'32.41"E)

Similarly, the Neukölln Town Hall was identified as a candidate despite its steeply pitched roof, which is practically unsuitable for standard green roof installations. These instances highlight that while the model captures general trends, specific architectural constraints—such as heritage status and complex roof geometry—still require manual verification.

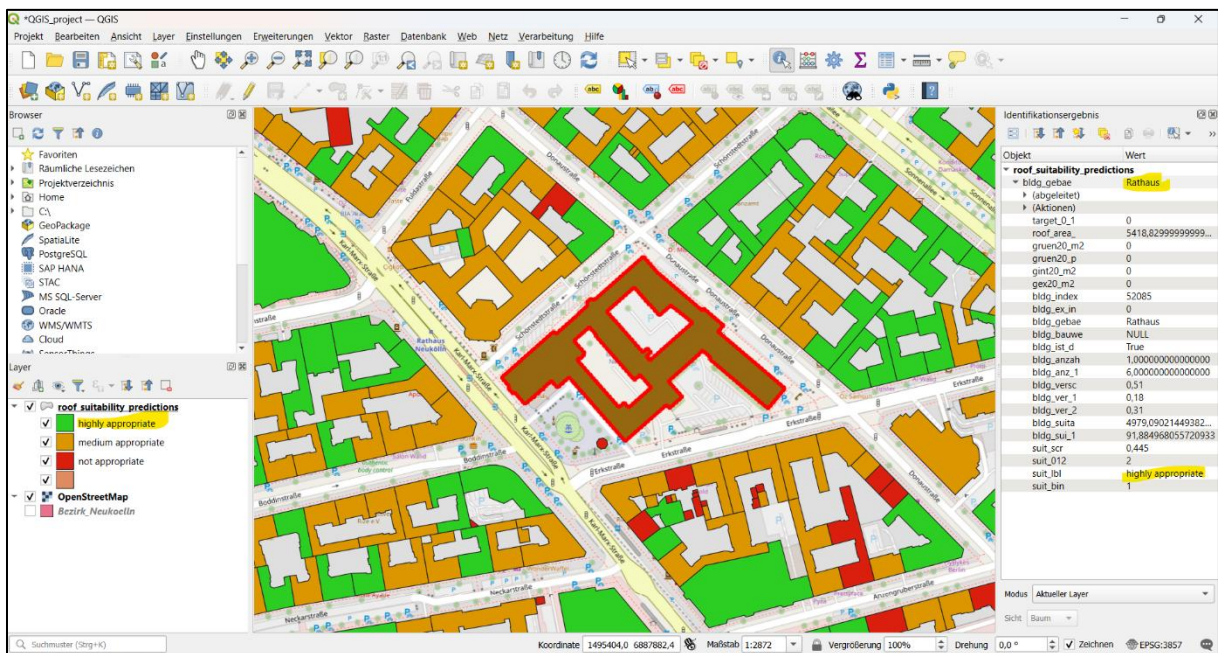


Figure 11: Townhall of Neukölln predicted as highly appropriate (in QGIS)

In comparison with Google Earth:

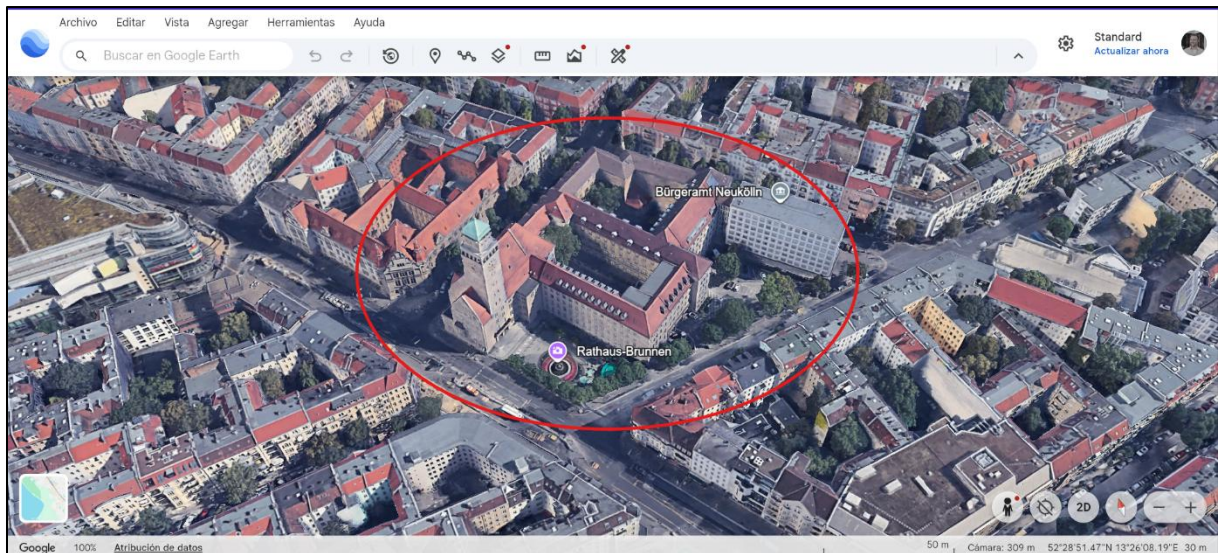


Figure 12: view of the townhall Berlin-Neukölln (Google Earth: 52°28'52.72"N 13°26'09.22"E)

5. Discussion

The results of this study demonstrate that predicting green roof suitability using open-access GIS data and machine learning is a viable, yet limited approach. While high AUC-ROC scores (~ 0.90) indicate a strong fundamental ability to rank buildings by suitability, several factors constrain the model's absolute predictive power.

5.1 Data limitations and Parameter selection

A primary limitation lies in the scope of the selected parameters. While roof slope, shading, and the number of stories serve as effective proxies for suitability, they do not fully capture a building's structural integrity. Future research should incorporate more granular data, such as building age or BIM-based structural information. Building age, in particular, could provide deeper insights into construction materials and static load capacities—factors that were only indirectly addressed via building height.

Furthermore, the preprocessing of parameters like shading could be intensified. Our current area-based categorization could be refined through a more sophisticated spatio-temporal analysis of light intensity to better assess diverse plant viability.

To refine these results, feature engineering could be employed to guide the decision tree more effectively. By creating custom columns that aggregate domain knowledge, we can weight specific factors that influence green roof suitability. For instance, building types could be scored based on their appropriateness: industrial buildings would receive a higher score than historical structures.

Similarly, environmental factors like solar exposure could be integrated; areas with minimal shading would be assigned higher values than those in full shade. By incorporating these scored features before training, the model can better account for architectural and environmental constraints, leading to more realistic and actionable predictions.¹¹

¹¹ Géron, A. (2019). *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems* (2nd ed.). O'Reilly Media

5.2 Methodological Challenges

The dataset suffers from a significant **class imbalance**, with existing green roofs accounting for less than 5% of Berlin's buildings. Despite implementing class-balancing techniques, this imbalance remains a major hurdle for Precision and the F1-Score. The resulting high number of false positives suggests that while the model is excellent at finding potential, it lacks the precision for immediate, unverified investment decisions.

Our "easy-to-handle" approach intentionally traded the complexity of deep learning and orthophotos for accessibility. However, this lacks the spatial context that image-recognition algorithms might provide. Additionally, alternative models like Gradient Boosted Trees (XGBoost) or Support Vector Machines might handle the non-linear relationships of urban geodata more effectively than the CART and Random Forest models used here.

6. Resume

Overall, the workflow provides a solid baseline for city-scale screening of green roof potential. Threshold tuning improved minority-class detection without retraining and should be retained for operational mapping. From a planning perspective, outputs should be interpreted as prioritization candidates rather than final decisions, with field verification as a required next step

Regarding the threshold tuning applied to the Neukölln district, the following conclusions can be drawn. The threshold can be adjusted to prioritize higher recall or a more conservative approach, depending on the specific objective. For the actual execution of green roof upgrades in Berlin, a conservative threshold is often more practical due to structural constraints, historical preservation requirements, and financial considerations.

However, since the primary goal of this project was to provide an initial assessment of green roof suitability across the city, adopting a broader threshold was more appropriate. This approach allowed us to capture a wider range of potential sites, offering valuable insights from an early-stage planning perspective and highlighting the overall potential for expansion.

Literature

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